



Pharmacokinetics

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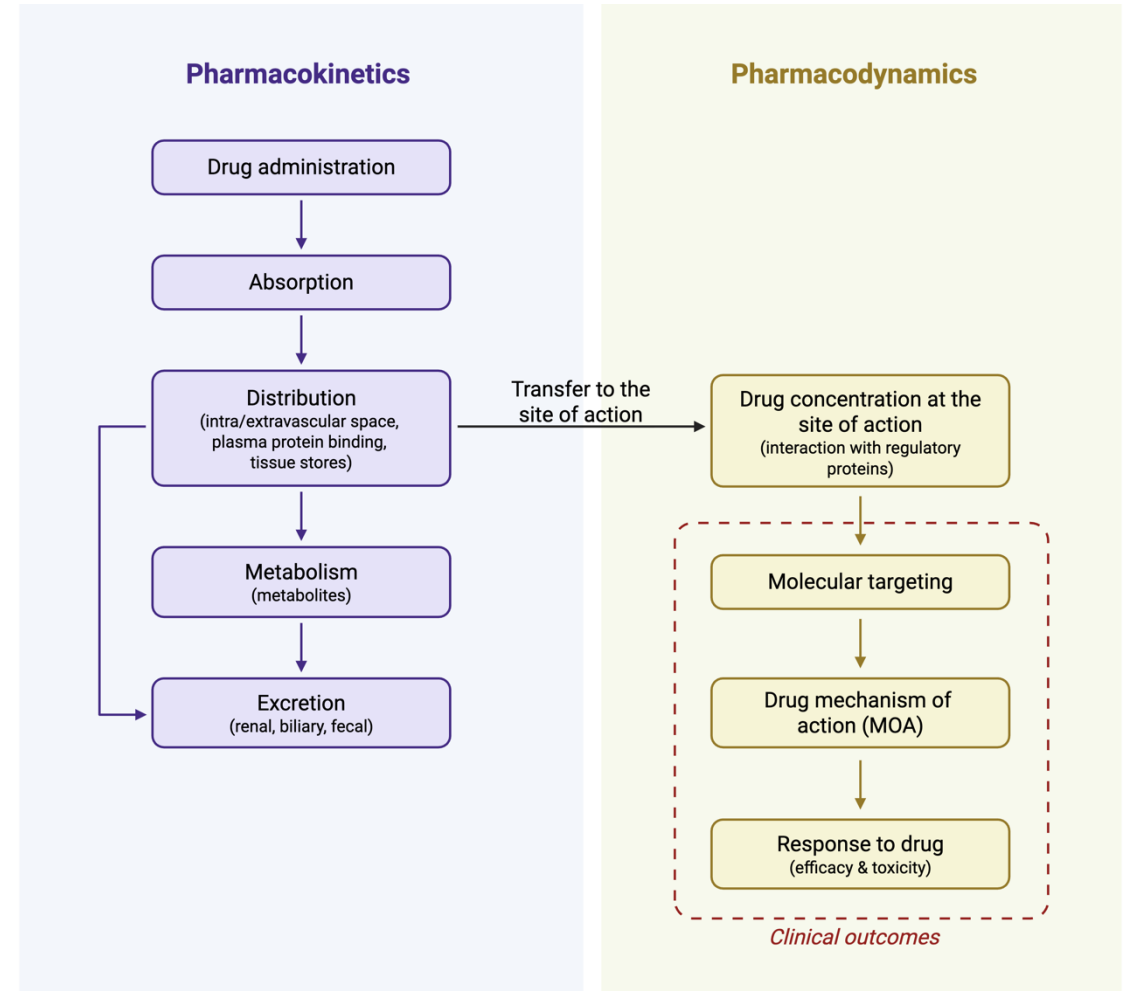
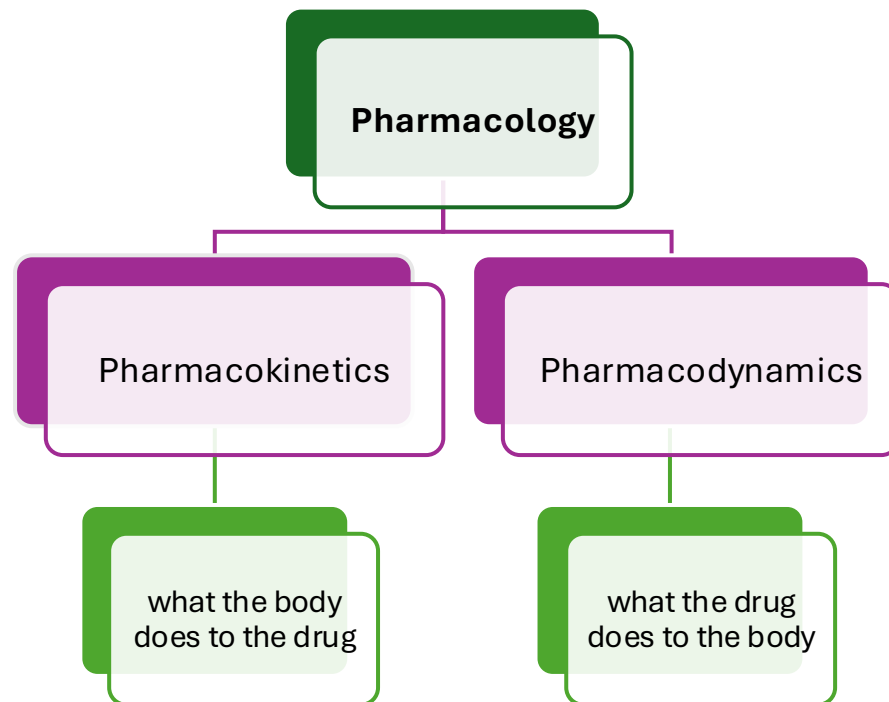
Learning objectives

- By the end of this lecture, students should be able to:
 1. Describe the major pharmacokinetic processes: absorption, distribution, metabolism, and excretion (ADME)
 2. Explain the clinical significance of key pharmacokinetic parameters, including bioavailability, volume of distribution, clearance, and half-life
 3. Perform basic pharmacokinetic calculations, including:
 1. Calculation of volume of distribution (V_d)
 2. Calculation of drug clearance (CL)
 3. Calculation of elimination half-life ($t_{1/2}$)
 4. Estimation of loading dose and maintenance dose
 4. Apply pharmacokinetic principles and calculations to simple clinical and exam-based scenarios

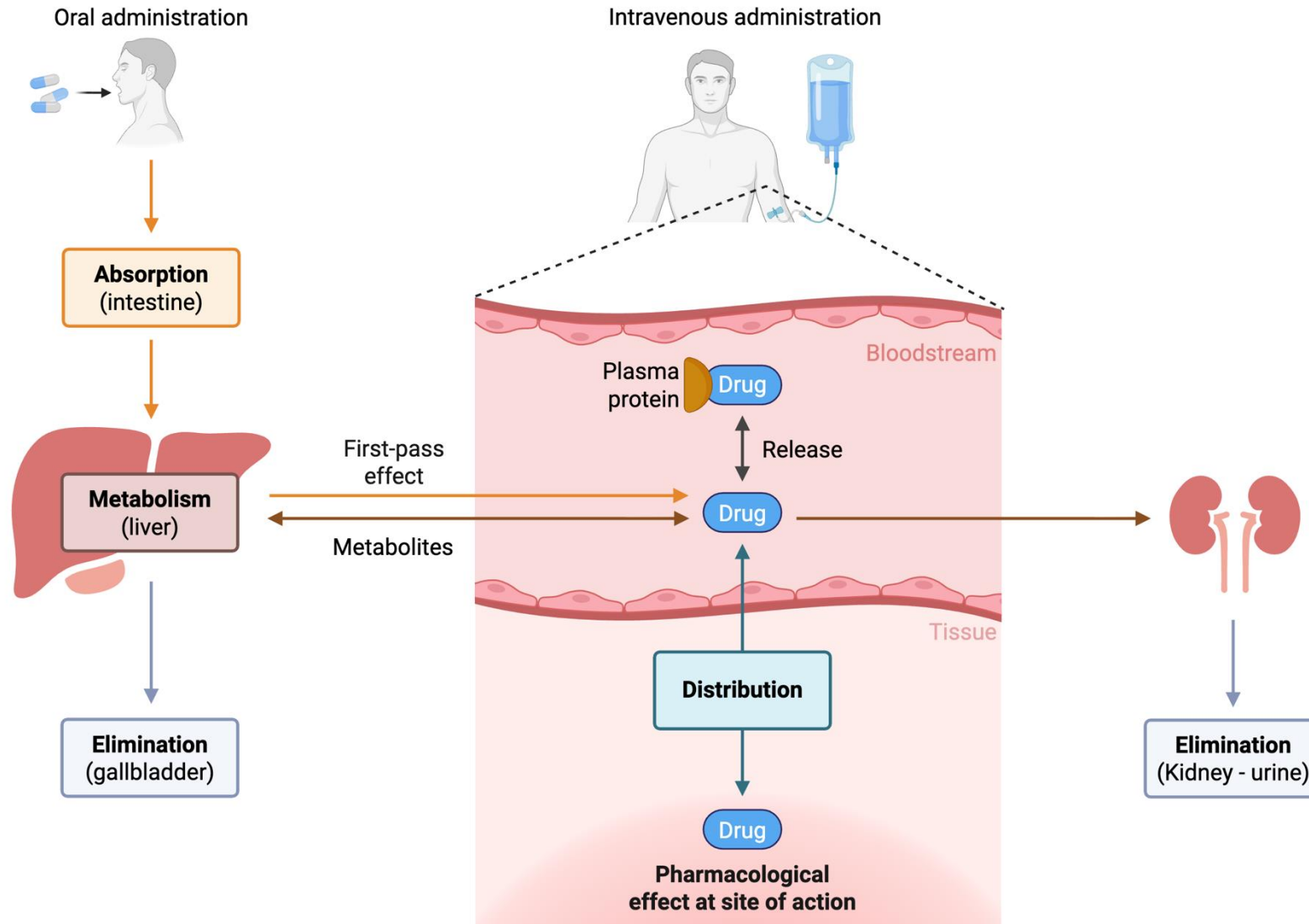
Introduction

WHAT IS PHARMACOLOGY?

The study of how drugs affect a biological system

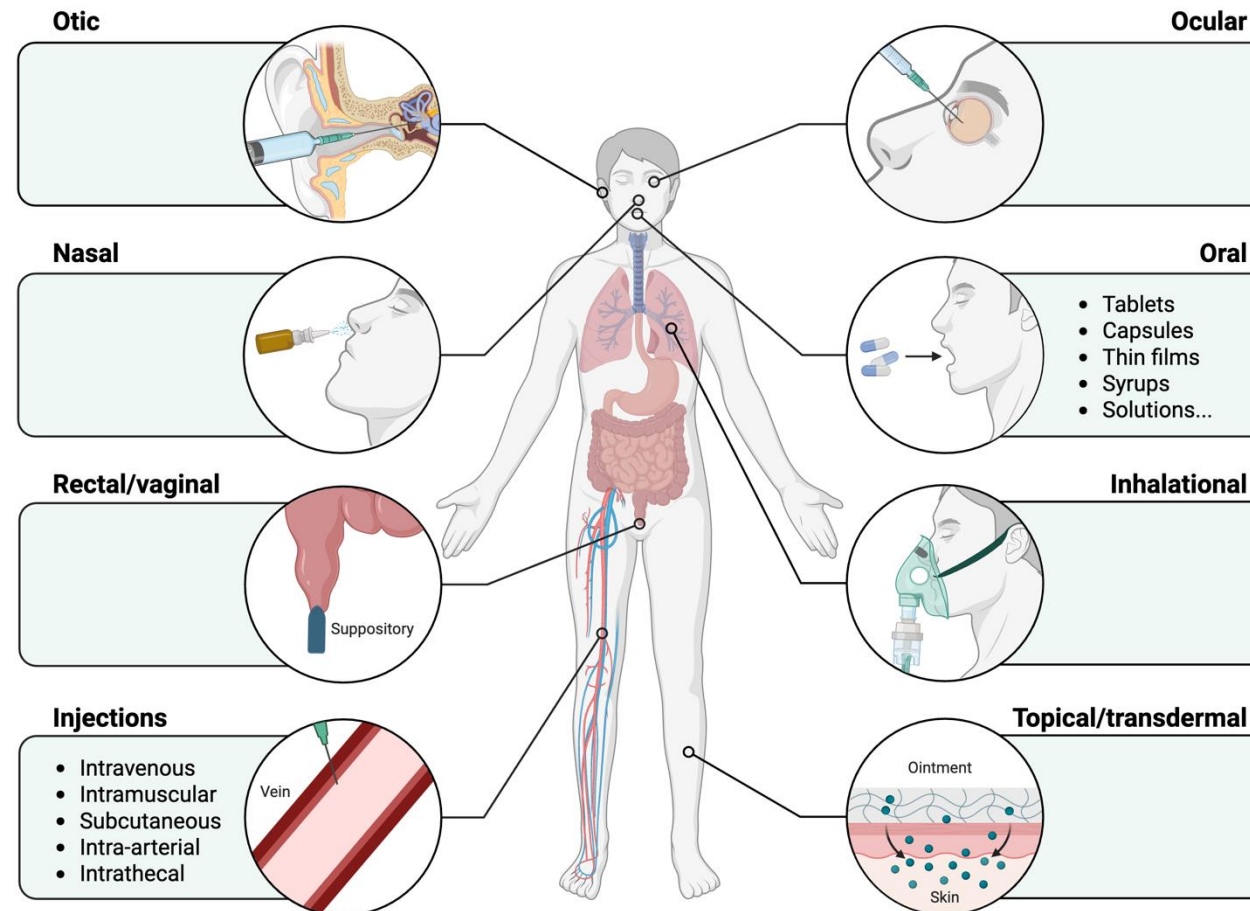


Pharmacokinetics Principals



Route of Drug Administration and PK

Drug Administration Routes



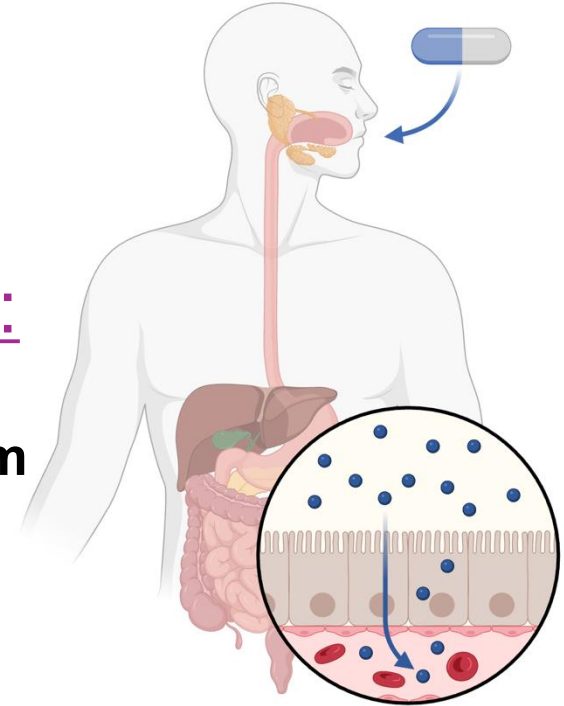
Absorption

Absorption is the transfer of a drug from its site of administration to the bloodstream.

Mechanisms of absorption of drugs from the GIT tract:

Depending on their chemical properties, drugs may be absorbed from the GI tract by

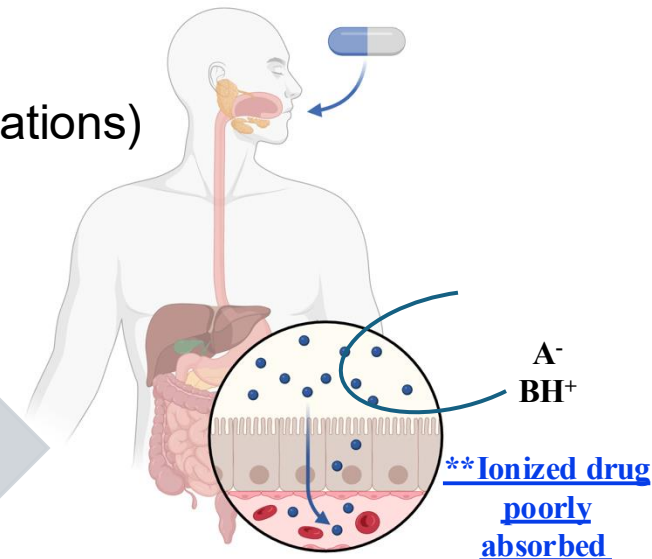
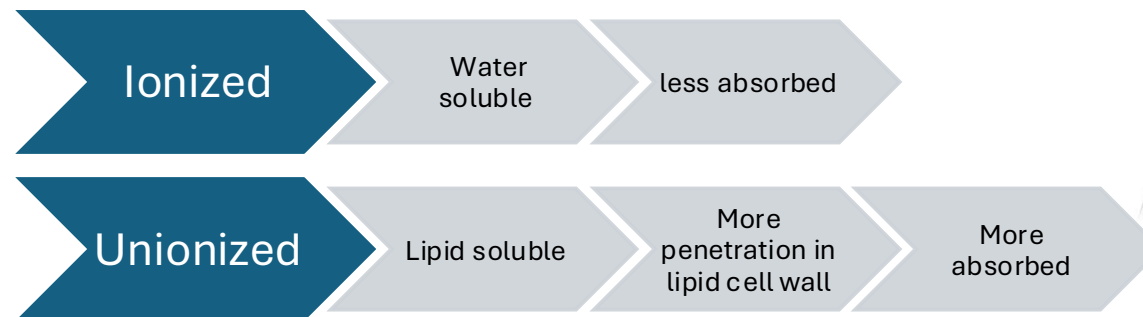
1. **Passive diffusion** (*Majority of drugs are absorbed by this mechanism*)
2. **Facilitated diffusion,**
3. **Active transport,**
4. **or Endocytosis**



Factors Affecting Absorption

Factors affecting absorption : (Drug-specific factors)

- **Route of administration.**
- **Dosage forms** (depending on particle size and disintegration, ease of dissolution).
(solution > suspension > capsule > tablet)
- **Molecular weight of drug:** Diffusion is most efficient with drugs that are small molecules.
- **Lipid solubility:** Drugs, which are usually somewhat **lipid-soluble** (fat-soluble), readily move across most biological membranes.
- **Degree of ionization:** Only **unionized form** is absorbable.
- **Drug solubility** (aqueous preparation better than oily, suspension preparations)



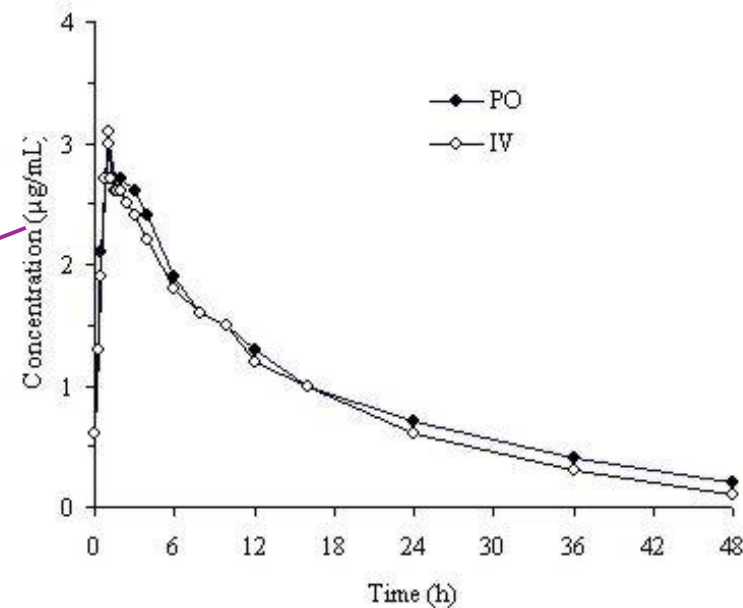
Factors Affecting Absorption

- **Factors Affecting Absorption** (patient-specific factors)
 - **Blood Circulation:** A patient who is in shock or cardiopulmonary arrest will need to have medications administered intravenously to achieve the desired response because of the reduced blood circulation in these situations.
 - **Contact time with the epithelial lining of the GI tract:** In people with a very rapid transit time through the GI tract, due to severe diarrhea for example, medication cannot be effectively absorbed.
 - **Drug-food or drug-drug interactions with other compounds present in the GI tract:** Specific foods or other drugs may bind a drug and prevent absorption. The interaction between tetracycline and dairy products or antacids is a good example of this effect.
 - Other examples:
 - High-fat → ↑ absorption of lipophilic drugs
 - Delays gastric emptying → delayed T_{max}
 - pH-altering drugs (antacids, PPIs) → change weak acid/base absorption
 - Enzyme/transporter modulators (P-gp inhibitors) → ↑ absorption

Bioavailability

- The **bioavailability** of a drug is the fraction (F) of the administered dose that reaches the systemic circulation.
- Bioavailability is defined as unity (or 100%) in the case of intravenous administration.
- For a drug administered orally, bioavailability may be less than 100% for two main reasons: **incomplete extent of absorption** across the gut wall and **first-pass effect** by the liver.

Amoxicillin shows very good bioavailability after oral administration

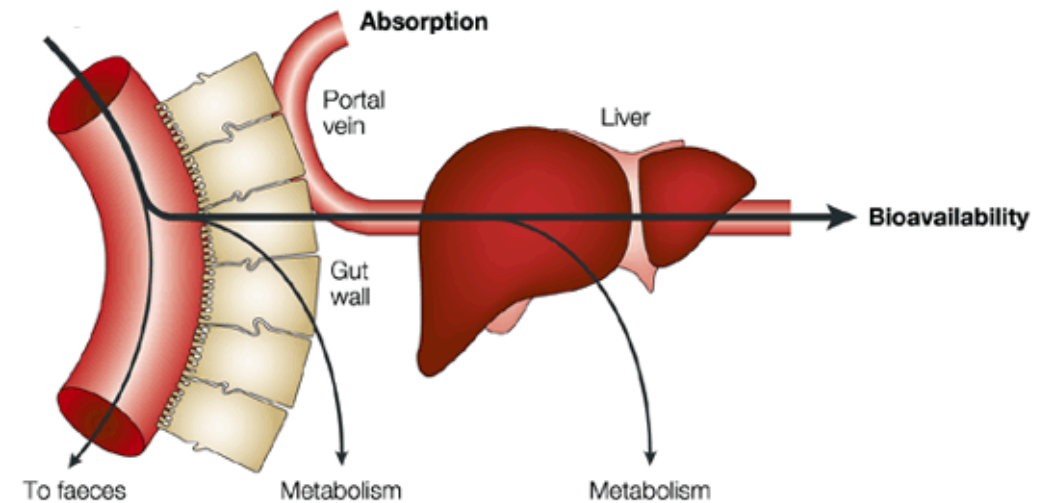


First Pass Metabolism

Definition: Drug metabolism before it reaches its site of action, occurring mainly in the liver and gut wall.

Clinical Impact:

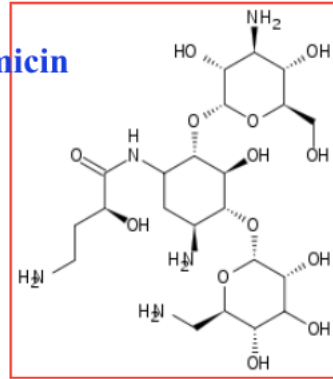
- Reduced oral effectiveness of some drugs (e.g., morphine)
- Higher oral doses may be needed to match IV effect (e.g., propranolol)
- Alternative routes avoid first-pass metabolism (e.g., sublingual nitroglycerin)



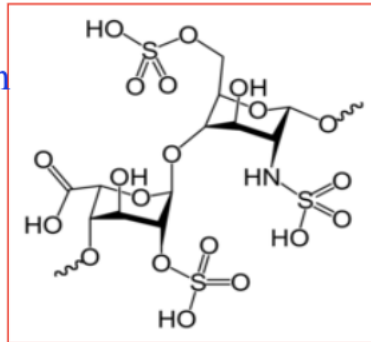
Explain the correlation between physicochemical properties and oral bioavailability

Not absorbed orally

Gentamicin



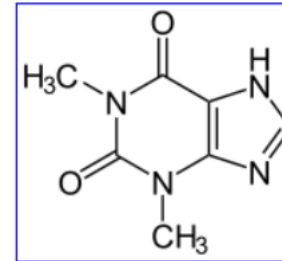
Heparin



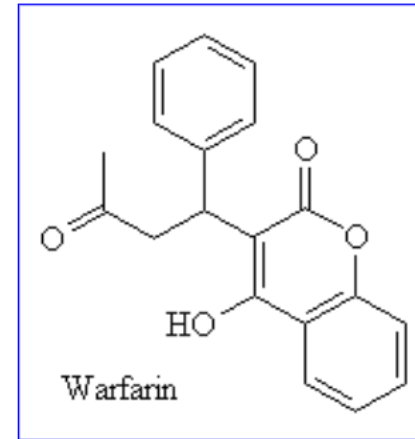
12000–15000 g/mol

Well absorbed orally

Theophylline

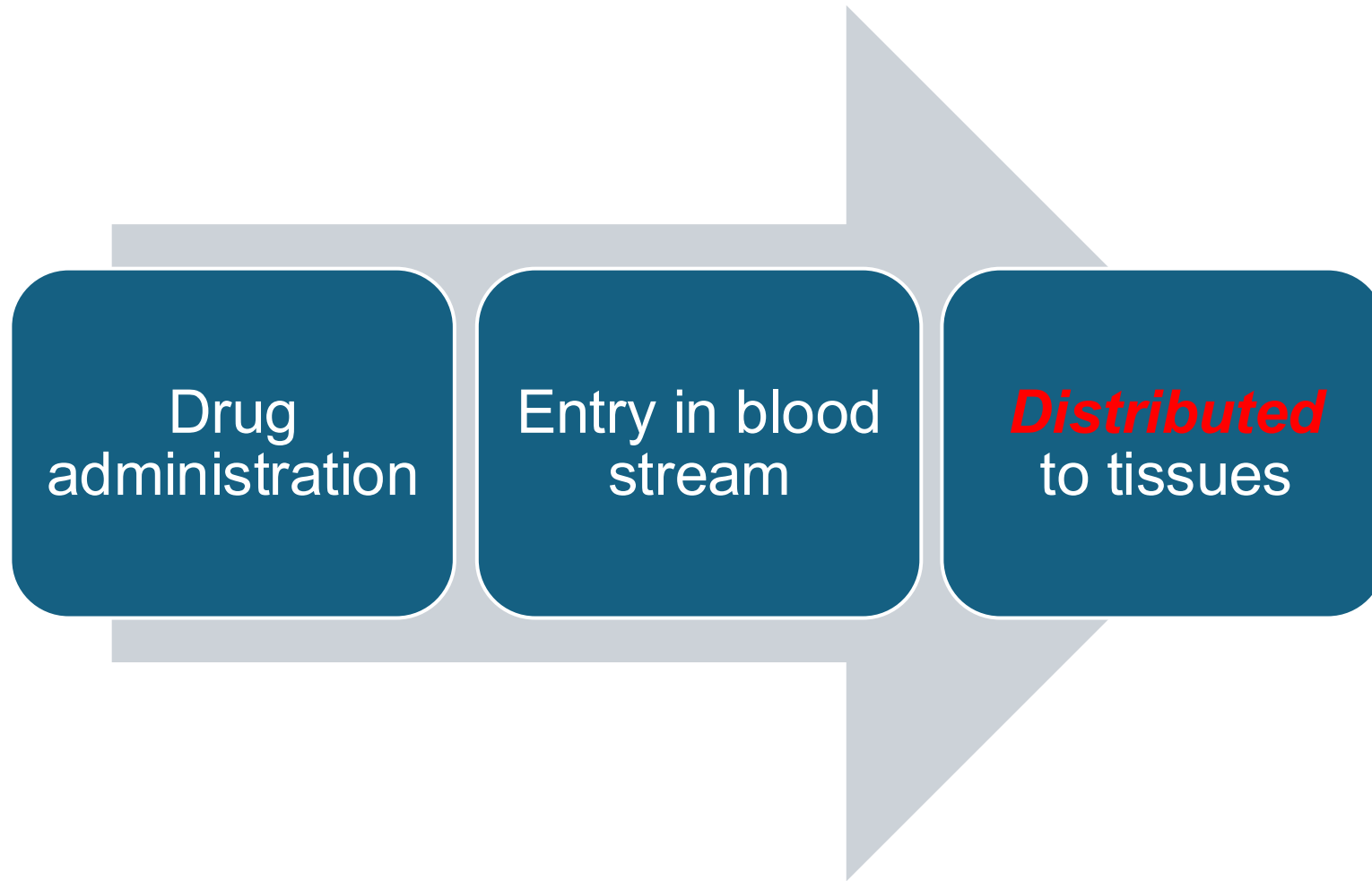


Warfarin



Distribution

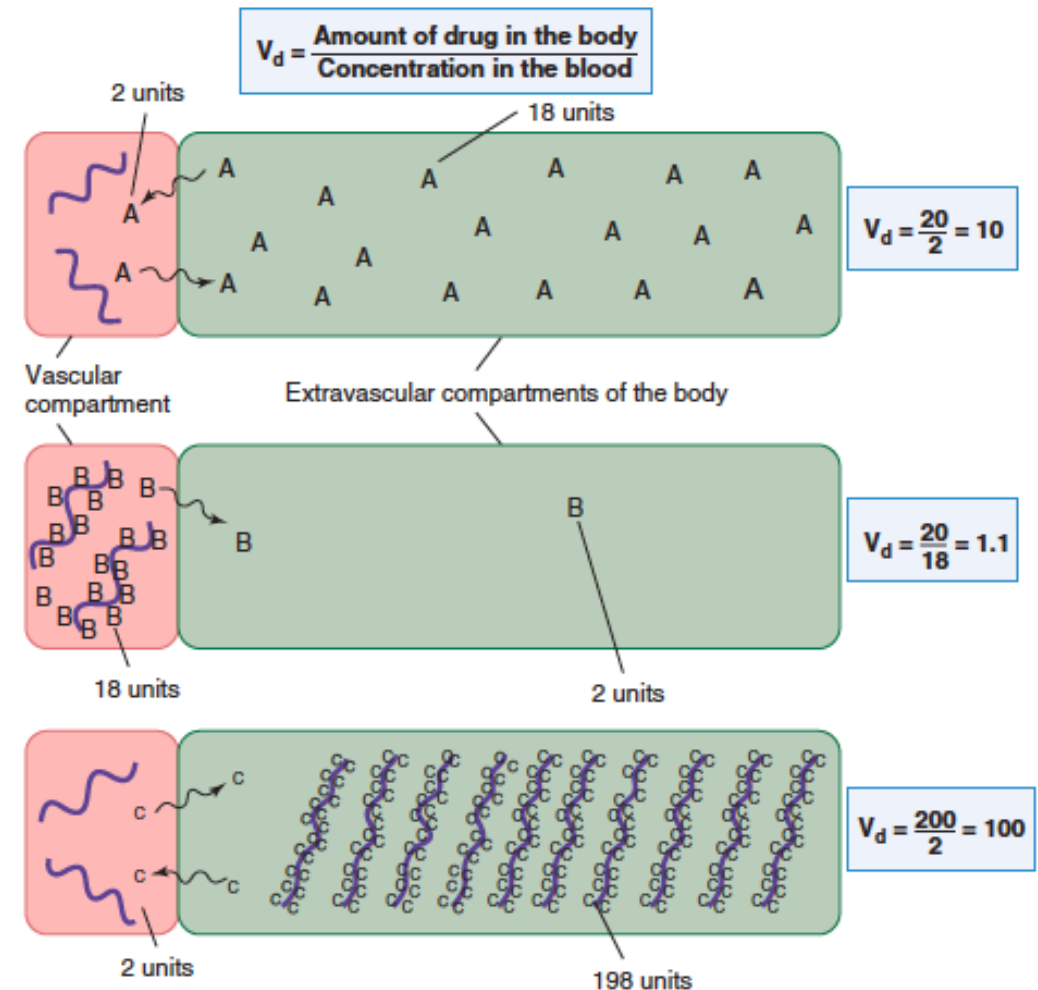
Distribution is the process by which a drug is carried to sites of action throughout the body by the blood- stream.



Distribution

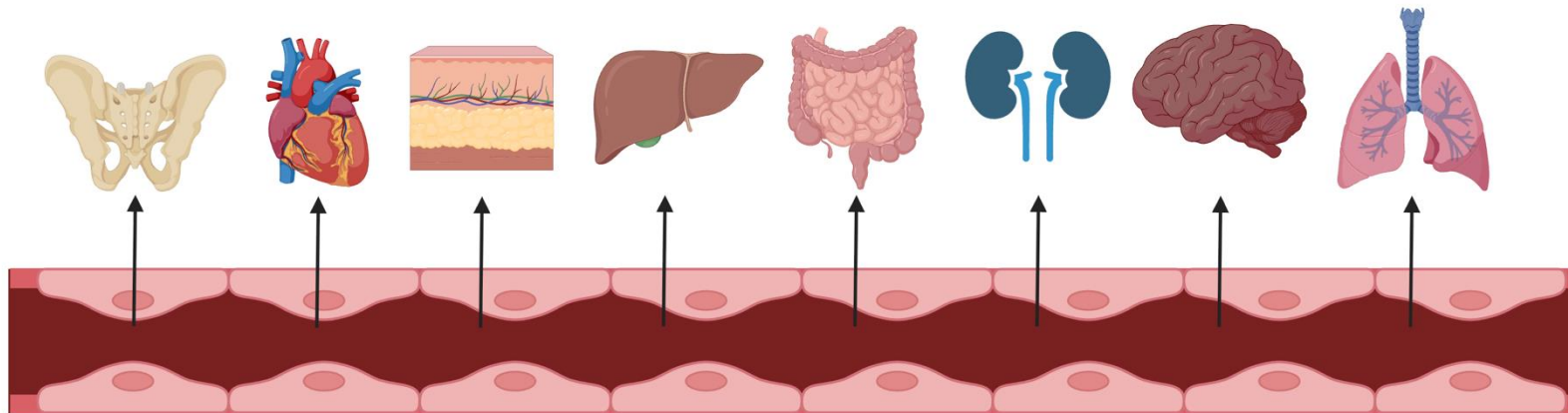
Volume of distribution (Vd) is the hypothetical volume needed to account for all of the drug in the body based on the serum concentration in a blood sample.

$$V_d(L) = \frac{\text{Total amount administered}}{\text{Plasma concentration}}$$



Factors affecting drug distribution

- **Chemical & physical properties**
- **Molecular weight & polarity** → affects tissue penetration
- **Lipid solubility** → lipophilic drugs cross membranes easily
- **Binding to blood components:**
 - **Plasma albumin** → strongly binds weak acidic drugs (e.g., warfarin, phenytoin), controls free drug fraction.
 - **α 1-acid glycoprotein & β -globulin** → bind certain basic drugs (e.g., propranolol, quinidine)
 - **Red blood cells (RBCs)** → some drugs partition into RBCs
 - **Lipoproteins** → bind highly lipophilic drugs (e.g., cyclosporine A)



Binding to plasma proteins

(Plasma protein binding, PPB)

- Drugs in blood exist in 2 forms: **free form** & plasma **protein-bound form**.
- Bound fraction is inactive, non-diffusible, cannot be metabolized or excreted.
- Free fraction is active, diffusible and can be metabolized and excreted.

Significance of binding of drugs to plasma proteins:

1. Acts as a reservoir.
2. Drug interactions: e.g., aspirin (higher affinity to plasma proteins) displaces warfarin (lower affinity to plasma proteins) → ↑ free fraction of warfarin (↑ activity) → bleeding.

Factors affecting drug distribution

Patient-related Factors

- **Age & body composition**

Neonates / elderly → ↑ or ↓ total body water, lean mass, fat → changes volume of distribution (V_d)

- **Physiological states**

e.g, Pregnancy: ↓ albumin, ↑ body water → ↑ free drug fraction

- **Blood flow & capillary permeability**

- High-flow organs (liver, kidney, heart) → rapid distribution
- Blood-brain barrier → restricts polar drugs

Factors affecting drug distribution

Disease-related Factors

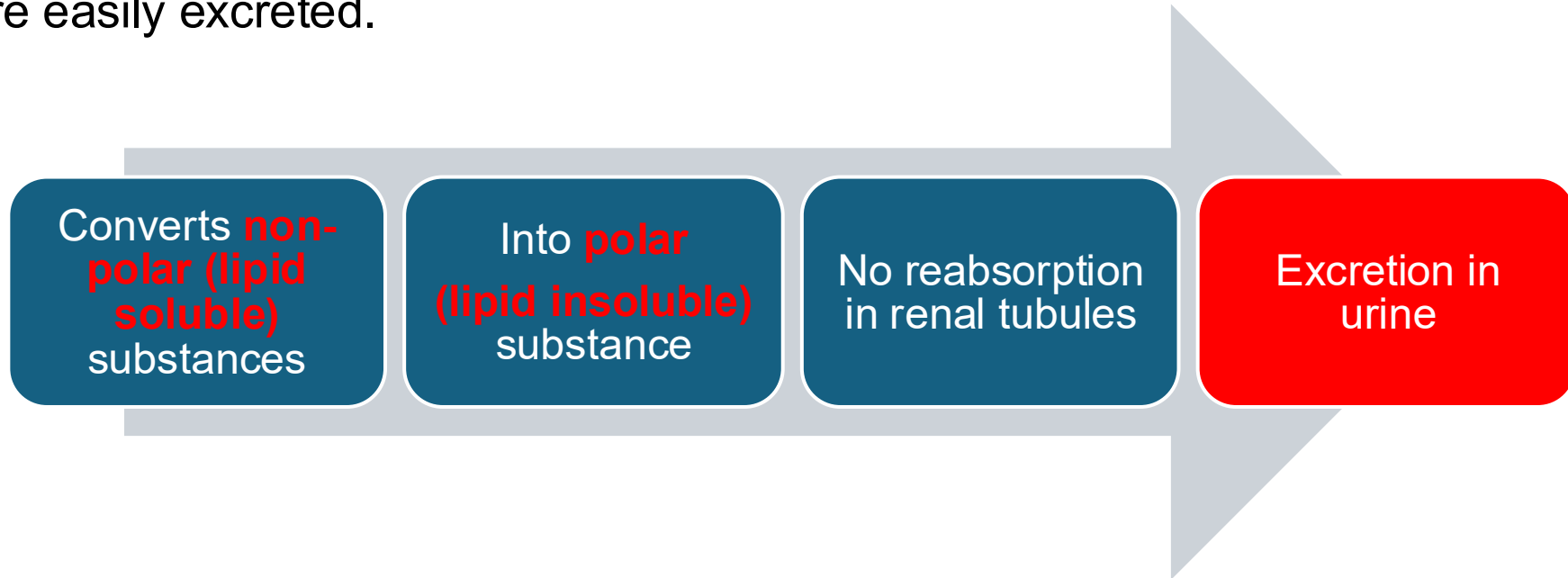
Disease	Effect on drug distribution
Liver disease (sever)	↓ plasma proteins → ↑ free drug fraction
Renal impairment / uremia	↓ albumin → ↑ free drug fraction
Diabetes	↑ free fatty acids displace drugs from albumin → ↑ free drug fraction
Cystic fibrosis	↑ blood volume → ↑ Vd for hydrophilic drugs

Clinical Impact of Altered Protein Binding

- **Protein binding is reversible & dynamic** → only free drug is pharmacologically active
- **Higher free drug levels** occur in:
 - Liver/renal disease (↓ albumin)
 - Drug interactions (competition for binding)
 - Neonates / pregnancy (↓ albumin)
- **Clinical Significance**
- Significant for **highly protein-bound drugs (>90%)**
- Small changes in binding → large changes in free drug → ↑ pharmacologic or toxic effect
- Drugs affected: phenytoin, warfarin, diazepam
- Monitor dose in neonates, pregnant women, or patients with altered protein levels

Metabolism or Biotransformation

- **Metabolism** is a process by which drugs may undergo chemical changes that allow them to be more easily excreted.



Key Concepts:

- Individual variation: genetically determined → affects metabolism
- Induction/Inhibition: altered by other drugs, foods, or environmental factors
- Primary site: liver, but extrahepatic metabolism may contribute

Metabolism or Biotransformation

- There are two types of metabolic processes drugs undergo in the liver:

1. In the first type of reaction (Phase I (Non-Synthetic) reactions) → drugs are made more polar through **oxidation-reduction reactions or hydrolysis**. These reactions use metabolic enzymes, most often those of the cytochrome P450 enzyme system, to catalyze the biotransformation.



Metabolism or Biotransformation

- There are two types of metabolic processes drugs undergo in the liver:

2. The second type of metabolism (Phase II (Synthetic) reactions) → involves conjugation reactions. In this type of reaction, the drug is joined with another substance, such as glucuronic acid, sulfuric acid, acetic acid, or an amino acid. Glucuronidation is the most common conjugation reaction. The result of conjugation is a more water-soluble compound that is easier for the kidneys to excrete.



Factors affecting drug metabolism

1. Genetic Factors

- **Polymorphisms** alter metabolism rate
 - Example: **Acetylation status** → fast, slow, normal (e.g., **isoniazid, INH**)

2. Liver Function

- **Advanced cirrhosis:** ↓ **Phase I reactions** (oxidation, reduction, hydrolysis) → slower metabolism of many drugs
- **Biliary obstruction:** impaired **Phase II metabolism & biliary excretion** → reduced elimination of drugs excreted in bile

3. Other Drugs

- **Enzyme inducers** → ↑ metabolism → ↓ drug levels (e.g., rifampin, carbamazepine)
- **Enzyme inhibitors** → ↓ metabolism → ↑ drug levels (e.g., ketoconazole, cimetidine)

4. Age (Extremes)

- **Elderly:** ↓ hepatic enzyme activity → slower metabolism
- **Infants < 6 months** (especially premature) → immature liver → ↓ clearance
- **Clinical impact:** drugs metabolized by liver → ↑ half-life, ↑ risk of accumulation/toxicity

Enzyme induction

- is a process in which a molecule (e.g. a drug) induces (i.e. enhances) the metabolizing capacity of a certain microsomal enzyme.
- **Stimulate** specific CYP e.g. CYP3A4
- **Enhance** Metabolism of certain drugs which are substrate for the same enzyme
- This leads to lower drug level → Reduce pharmacological effect (possibly therapeutic failure)
- Example: Tobacco smoking induces CYP1A2 (example substrates are clozapine/olanzapine)

Enzyme inhibition

- It is the decrease of the rate of metabolism of certain drug caused by another drug .
- This will lead to the increase of the concentration of the target drug and leading to the increase of its toxicity
- Inhibition of the enzyme may be due to the competition on its binding sites , so the onset of inhibition is short may be within 24h.
- NB: Inhibition = decrease of drug metabolism (certain drugs)

Elimination

- The complete removal of the drug from the body is referred to = **Elimination**.
- Drugs must be sufficiently **polar** to be eliminated from the body.
- Removal of drugs from the body occurs via a number of routes; the most important is elimination through the kidney into the urine.
 - Other routes of drug removal are:
 - volatile agents via lungs
 - bile
 - intestine
 - milk in nursing mothers

N.B: Patients with renal dysfunction may be unable to excrete drugs and are at risk for drug accumulation and adverse effects.

Renal excretion of drugs

1-Filtration

- Passive process (Pressure driven); Small molecules
- Most proteins not filtered. Drugs which are extensively protein bound will also not be filtered.

2-Active secretion

- require Energy ; Two separate mechanisms for acids & bases
- Saturable; Possible interactions

3-Re-absorption

- Depends on urine flow & PH of urine (state of drug ionization & lipid solubility)
- In case of toxicity of certain drugs insure adequate urine flow & appropriate pH that insure ionization

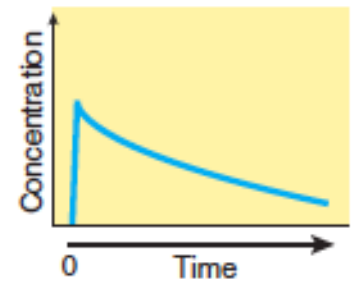
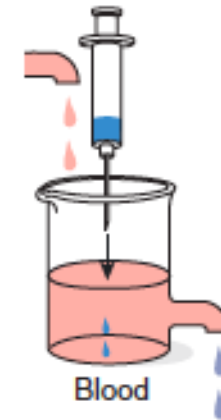
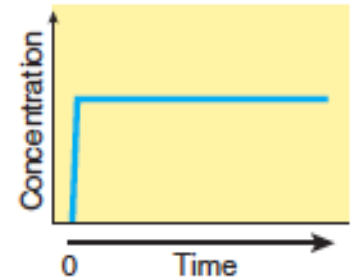
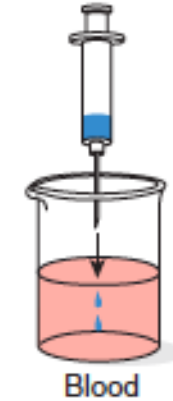
Elimination

- Parameters of elimination

1. Clearance
2. Elimination half life ($t_{1/2}$)
3. Kinetics orders

- Clearance (CL) relates the rate of elimination to the plasma concentration.

$$\text{Clearance (CL)} = \frac{\text{Rate of elimination}}{\text{Plasma concentration (Cp)}}$$

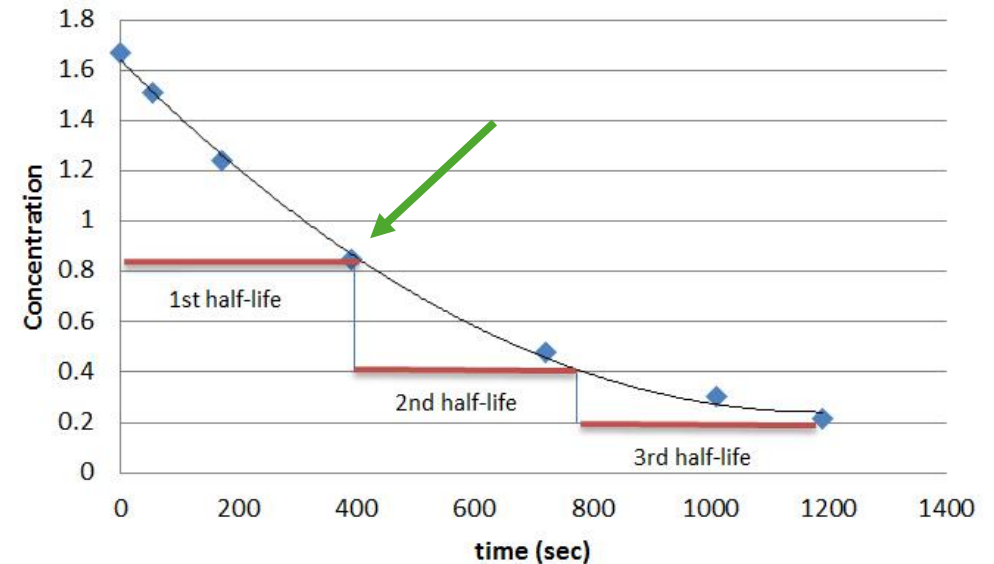


2. Half Life ($t_{1/2}$)

- Half-life ($t_{1/2}$) is the time required for the amount of drug in the body or blood to fall by 50%.
- Half-life ($t_{1/2}$) will depend on both the volume of distribution and the clearance:

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

- The half-life determines the rate at which blood concentration rises during a constant infusion and falls after administration is stopped.



2. Half Life ($t_{1/2}$)

- 1st $t_{1/2}$ = 50% drug eliminated
- 2nd $t_{1/2}$ = 75% (50 + 25) eliminated
- 3rd $t_{1/2}$ = 87.5% (50+25+12.5) eliminated
- 4th $t_{1/2}$ = 93.75% (50+25+12.5+6.25) eliminated
- **So 4-5 half lives are required for complete elimination.**

3. Kinetics Orders

A. First order kinetics (most drugs)	B. Zero order Kinetics e.g., Phenytoin, ethanol
1. Rate of elimination of a drug is proportional to its concentration (Clearance (Cl) $\propto$ C) i.e., a constant <u>fraction (ratio)</u> of the drug is eliminated per unit time.	• Rate of elimination of a drug is constant i.e., a constant <u>amount</u> of the drug is eliminated per unit time.
2. $t_{1/2}$: Constant.	• $t_{1/2}$: not constant
3. $T_{ss} = 4-5 t_{1/2}$ It is time needed to reach Concentration steady state (C_{ss})	No C_{ss} is reached (overshooting of concentration)
4. Modest changes in dose or bioavailability are usually tolerated.	• Modest changes in dose or bioavailability $\rightarrow$ toxicity.

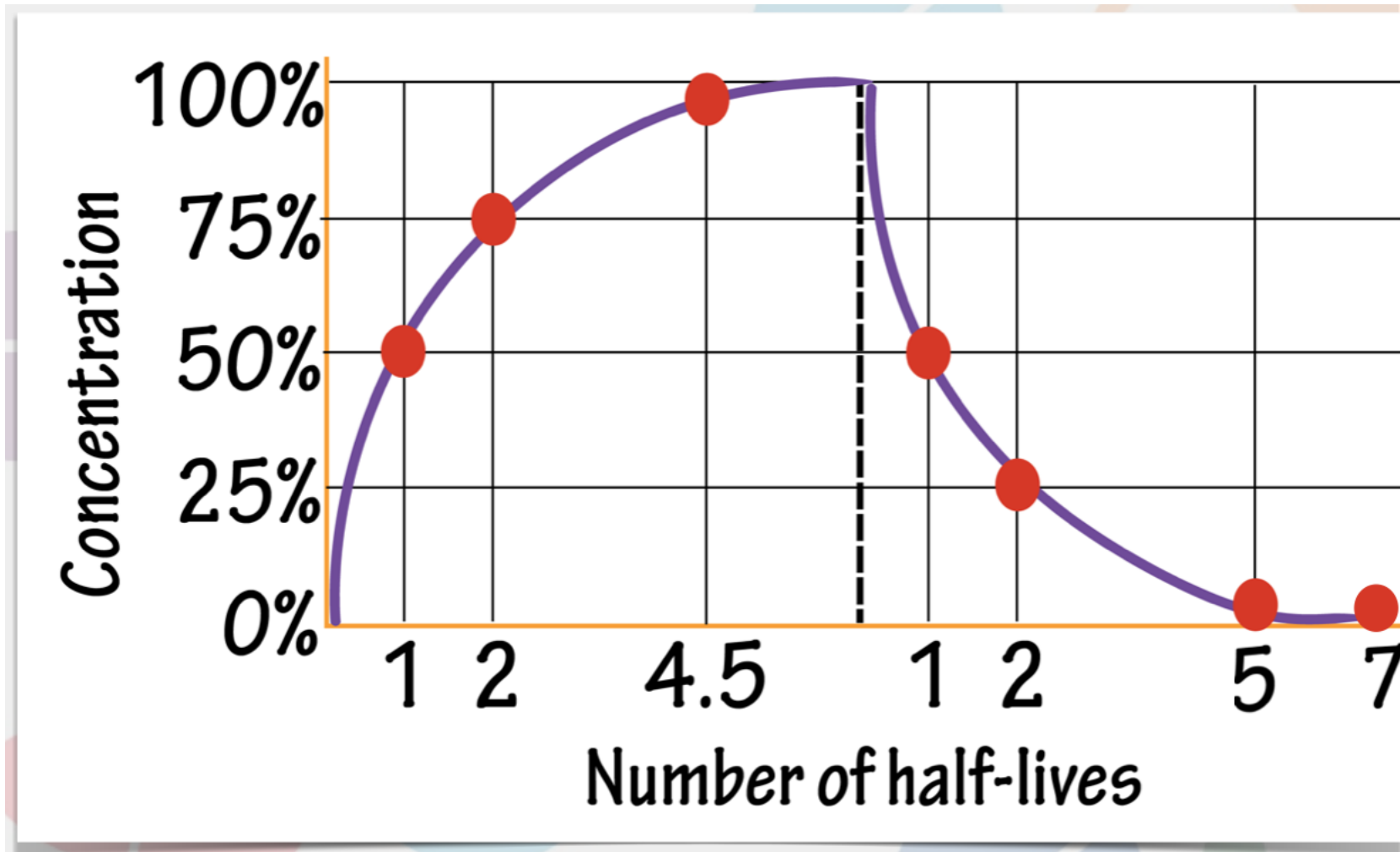
Pharmacokinetics Calculations



Value of pharmacokinetic principles for future medical doctors

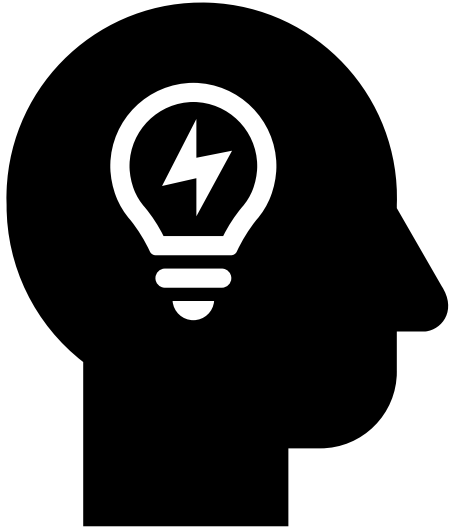
1. The choice of **loading and maintenance dose**.
2. The decision on the frequency and the route of administration.
3. Helps in predicting the rate and extent of drug accumulation.
4. Aids in estimating the effect of dose changes.
5. Identifying and anticipating drug interactions.
6. Identifying patient and disease factors that could alter clinical response.
7. Interpreting drug concentrations in serum or plasma.

Understanding the steady state



Rate of attainment of steady-state concentration of a drug in the plasma.

4-5 half-lives are required for reaching the steady state or complete elimination of the drug.



- A delay in achieving the desired plasma levels of drug may be clinically unacceptable.
- Therefore, a "loading dose" of drug can be injected as a single dose to achieve the desired plasma level rapidly, followed by an infusion to maintain the steady-state (maintenance dose).

Loading Dose

- From the equation of Volume of distribution (V_d) :

$$V_d = D \div C$$

where C = the plasma conc. of the drug.

and D = the total amount of drug in the body.

We can calculate the dose needed to achieve a certain peak plasma concentration:

$$\text{Loading Dose (LD)} = (V_d) \times (C_p)$$

Example

Assuming a one-compartment model, what would be the loading dose to achieve a target plasma concentration of 25 mg/L of a drug whose volume of distribution is 100 L?

- A. 250 mg.
- B. 750 mg.
- C. 1500 mg.
- D. 2500 mg .

$$\text{Loading dose} = V_d \times C_{ss}$$

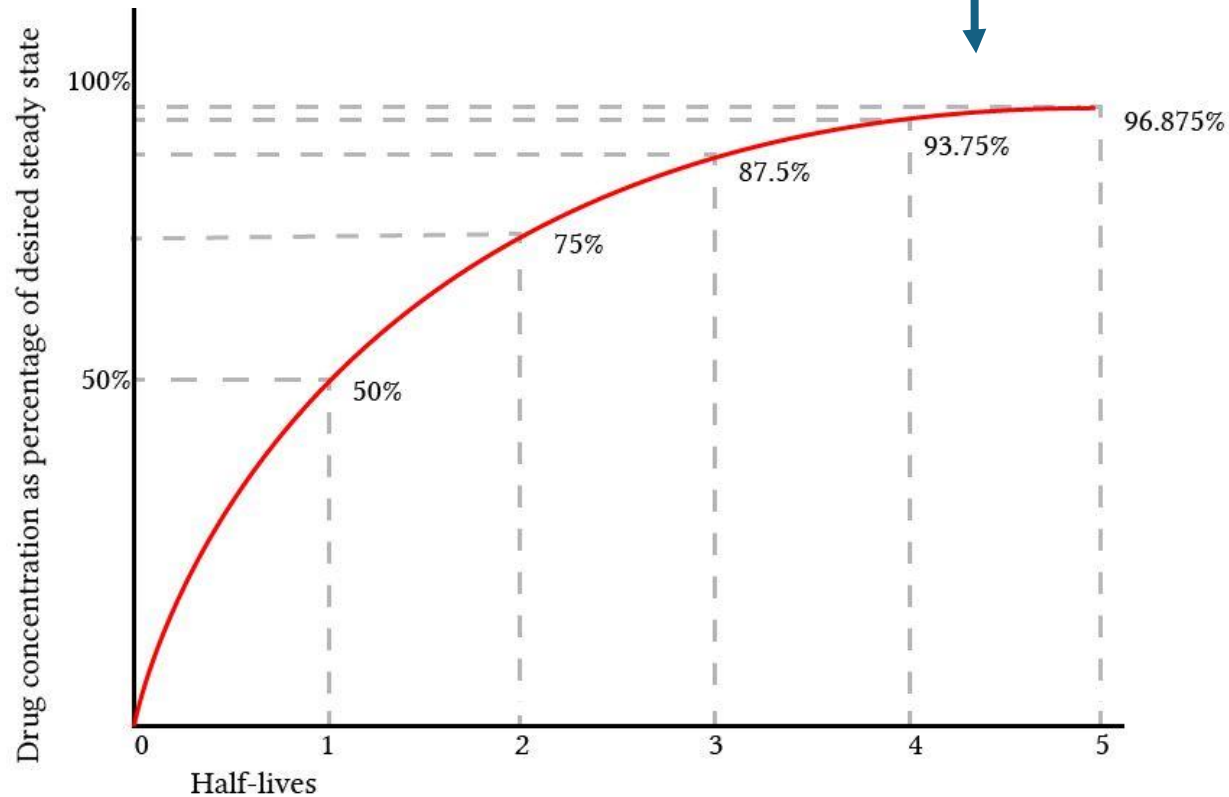
Example

What is the volume of distribution (V_d) of a drug whose concentration in the plasma is $2 \mu\text{g/mL}$ and the total amount of drug in the body is 42 mg ?

- A. 2.4 L
- B. 4.2 L
- C. 18 L
- D. 21 L

Maintenance Dose

Steady state concentration
of a drug occurs when
Rate of administration = Rate of elimination



$$R_0 \text{ (Maintenance dose or infusion rate)} = C_{ss} \times CL$$

C_{ss} = the steady state concentration of the drug.

Example

Suppose you want to give a drug by I.V. infusion at a dosing rate that would provide a target concentration of 20 mg/L. The clearance of the drug is 4 L/hr.

Which of the following would be the best dosing rate to use?

- A. 20 mg/hr
- B. 40 mg/hr
- C. 60 mg/hr
- D. 80 mg/hr

$$R_o = C_{ss} \times CL_t$$